



COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

Complex Engineering Problem (CEP) - Proposal Submission Form
Microprocessor Systems and Interfacing Lab (CPE-342)

Project Title	
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Range of Complex Problem Solving (According to PEC Manual):

	Name	Registration No.
1.		
2.		
3.		

Project Description: [Provide a brief description of what you are building. Explain the problem you are solving, the application domain (e.g., environmental monitoring, motor control, smart home, etc.), and the key functionality of your embedded system. This should be 150-200 words.]

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Project Objectives: [List 2-5 specific, measurable objectives that your project aims to achieve. Each objective should be clear and achievable within the project timeline.]

Objective	Description
1.	
2.	
3.	
4.	

System Block Diagram: [Insert a block diagram showing the major components of your system, their interconnections, communication protocols, and data flow. The diagram should clearly show the STM32 microcontroller, all peripherals, sensors, actuators, displays, and their interface connections. A handwritten diagram will work fine]

Expected Outcomes: [Describe what you expect to achieve by the end of this project. What will the working system demonstrate? What measurements or validations will you perform?]

Alignment with Sustainable Development Goals (SDGs): [Identify which UN SDG(s) your project aligns with.]

- ☐ SDG 7 (Affordable and Clean Energy)
- ☐ SDG 9 (Industry, Innovation, and Infrastructure)
- ☐ SDG 11 (Sustainable Cities and Communities)

Declaration:

We, the undersigned, declare that this proposal represents our own original work and that we understand the requirements and expectations of the Complex Engineering Problem (CEP) project for CPE-342. We understand that in case of plagiarism we will be awarded an F-grade in the course.

Sign and Date:

Name:	Name:	Name:
Sign:	Sign:	Sign: